Rohan Verma

Embedded Systems & IoT Firmware Engineer

Roll No: 22EC019 | Department: Electronics & Communication | CGPA: 8.65 / 10.0

Email: rohan.v@college.edu | Phone: +91 98777 88990 | LinkedIn: linkedin.com/in/rohan-verma-embedded

PROFESSIONAL SUMMARY

Hardware-software integration engineer with hands-on expertise in microcontroller programming (STM32, ESP32), FreeRTOS real-time scheduling, low-power IoT telemetry, and embedded C/C++.

CORE TECHNICAL SKILLS

- **Languages:** Embedded C, C++20, Python, Assembly
- **Hardware & MCUs:** ARM Cortex-M, STM32, ESP32, Arduino, Raspberry Pi, Nordic nRF
- **Protocols:** I2C, SPI, UART, CAN, MQTT, BLE, CoAP, Zigbee
- **Tools & RTOS:**FreeRTOS, Keil uVision, STM32CubeIDE, Logic Analyzer, KiCad

WORK EXPERIENCE & INTERNSHIPS

Firmware Engineering Intern | *SenseTech Instruments* Jan 2024 - Jun 2024

- Programmed low-power sensor drivers on ESP32 MCUs operating with FreeRTOS task queues.
- Extended sensor battery operating lifetime by 45% utilizing deep-sleep modes and burst telemetry.

FEATURED TECHNICAL PROJECTS

Industrial IoT Predictive Telemetry Node *(Embedded C, ESP32, MQTT, AWS IoT Core)*

- Deployed vibration and thermal monitoring nodes streaming live machine health metrics to AWS.

Autonomous Obstacle-Avoiding Robotics Rover *(C++, ROS, Raspberry Pi, LiDAR)*

- Implemented SLAM navigation algorithms and sensor fusion combining ultrasonic and LiDAR readings.

EDUCATION & ACADEMIC BACKGROUND

Bachelor of Technology in Electronics & Communication • Delhi Technological University 2022 - 2026

CGPA: 8.65 / 10.0 | Relevant Coursework: Microprocessors & Microcontrollers, Digital Signal Processing, Embedded Systems

CERTIFICATIONS & HONORS

- ARM Embedded Systems Architecture Certificate
- Embedded Linux System Development (The Linux Foundation)